

MAGNETISM AND MATTER

CHAPTER FIVE CLASS - XII

MANISH JOSHI

D.A.V. LOHAGHAT

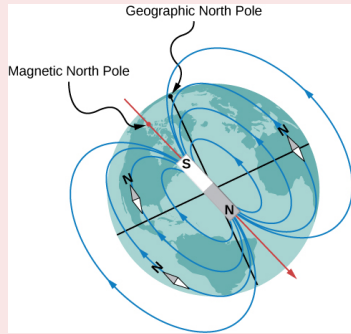
October 4, 2024

Learning Objectives:

- Bar magnet, Magnetic field lines
- Bar magnet as an equivalent solenoid (qualitative treatment only).
- Torque on a magnetic dipole (bar magnet) in a uniform magnetic field (qualitative treatment only).
- Magnetic field intensity due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis (qualitative treatment only).
- Magnetic properties of materials- Para-, dia- and ferro-magnetic substances with examples.
- Magnetization of materials.
- Effect of temperature on magnetic properties.

Introduction

Some of the commonly known ideas regarding magnetism are:



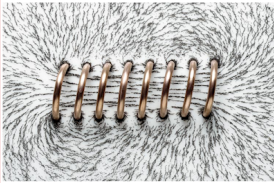
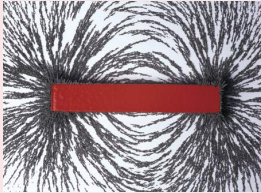
- The **earth behaves as a magnet** with the magnetic field pointing approximately from the geographic south to the north.
- When a **bar magnet is freely suspended**, it **points in the north-south direction**. The tip which points to the **geographic north** is called the **north pole** and the tip which points to the **geographic south** is called the **south pole** of the magnet.

Introduction

Some of the commonly known ideas regarding magnetism are:

- Magnetic poles **repel** if they are **alike** (both N or both S), they **attract** if they are **opposite** (one N and the other S)
- **We cannot isolate the north, or south pole** of a magnet. If a bar magnet is broken into two halves, we get two similar bar magnets with somewhat weaker properties. **Unlike electric charges**, isolated magnetic north and south poles known as **magnetic monopoles do not exist**.
- It is **possible** to **make magnets** out of **iron and its alloys**.

THE BAR MAGNET AND MAGNETIC FIELD LINES



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- By examining the **pattern of iron filings** sprinkled on a sheet of glass placed over a short **bar magnet**.
- The **pattern** of iron filings **suggests** that the **magnet has two poles. similar to the positive and negative charge** of an electric dipole.
- A **similar pattern** of iron filings is **observed** around a current carrying **solenoid**.

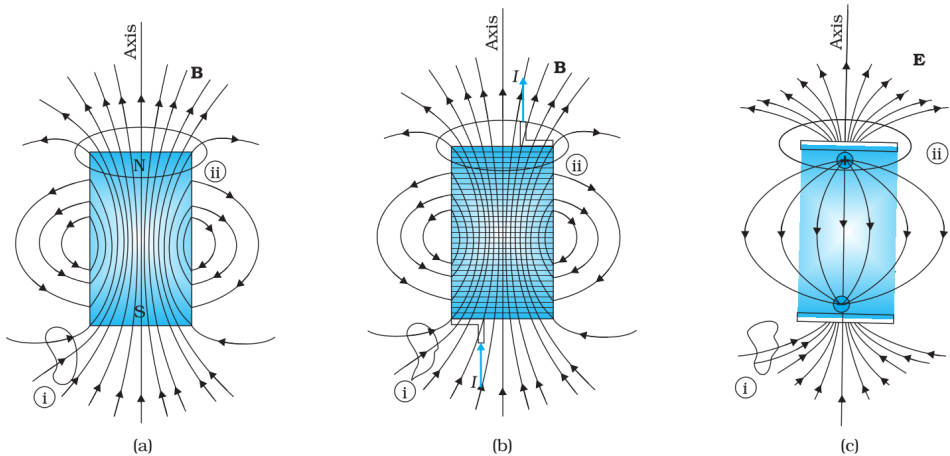


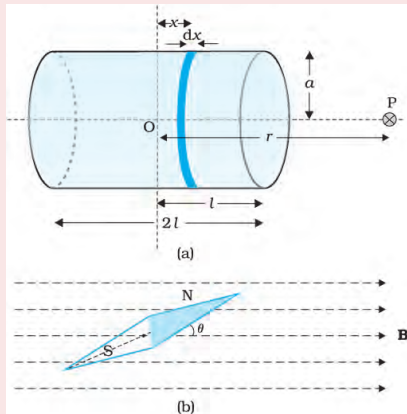
FIGURE 5.3 The field lines of (a) a bar magnet, (b) a current-carrying finite solenoid and (c) electric dipole. At large distances, the field lines are very similar. The curves labelled (i) and (ii) are closed Gaussian surfaces.

THE BAR MAGNET AND MAGNETIC FIELD LINES

The **magnetic field lines** are a **visual and intuitive realisation** of the **magnetic field**. Their **properties** are:

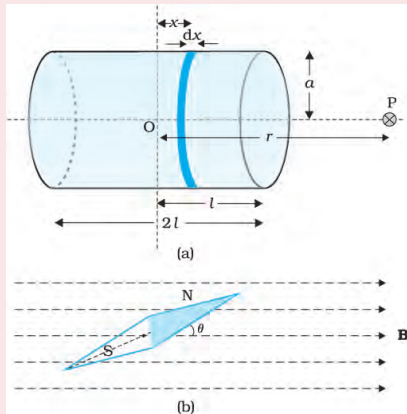
- The **magnetic field lines** of a magnet (or a solenoid) **form continuous closed loops**. This is **unlike the electric dipole** where these field lines **begin from a positive charge** and **end on the negative charge or escape to infinity**.
- The **tangent to the field line** at a given point **represents the direction of the net magnetic field B** at that point.
- The **larger the number of field lines crossing per unit area**, the **stronger is the magnitude of the magnetic field B** .
- The **magnetic field lines do not intersect**, for **if they did**, the **direction of the magnetic field would not be unique** at the point of intersection.

Bar magnet as an equivalent solenoid



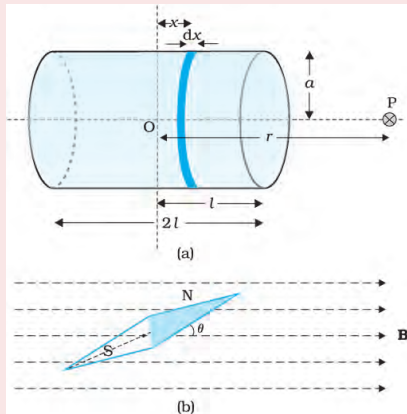
- The **resemblance of magnetic field lines for a bar magnet and a solenoid suggest** that a **bar magnet may be thought of as a large number of circulating currents in analogy** with a solenoid.
- To make this analogy more firm **we calculate the axial field of a finite solenoid.**
- We shall **demonstrate that at large distances this axial field resembles** that of a **bar magnet.**

Bar magnet as an equivalent solenoid



- Let, n be the **number of turns per unit length**.
- Let solenoid **length** be $2l$ and **radius** a .
- We want to **evaluate** the **axial field** at a point **P**, at a **distance** r from the **centre** **O** of the solenoid.
- **consider** a **circular element** of **thickness** dx of the solenoid **at a distance** x from its centre. It **consists** of ndx turns. Let I be the **current** in the solenoid.

Bar magnet as an equivalent solenoid



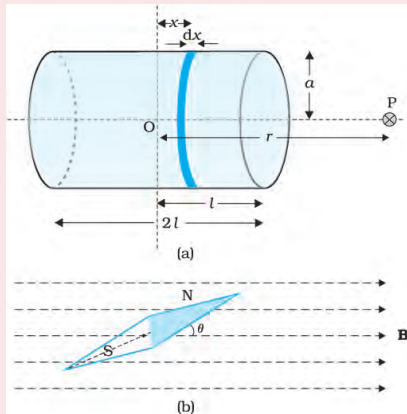
- The **magnitude of the field at point P** due to the **circular element** is

$$dB = \frac{\mu_0 l a^2 n \, dx}{2 [(r - x)^2 + a^2]^{3/2}}$$

- The magnitude of the total field is obtained by integrating from $x = -l$ to $x = +l$. Thus,

$$B = \frac{\mu_0 l a^2 n}{2} \int_{-l}^{+l} \frac{dx}{[(r - x)^2 + a^2]^{3/2}}$$

Bar magnet as an equivalent solenoid



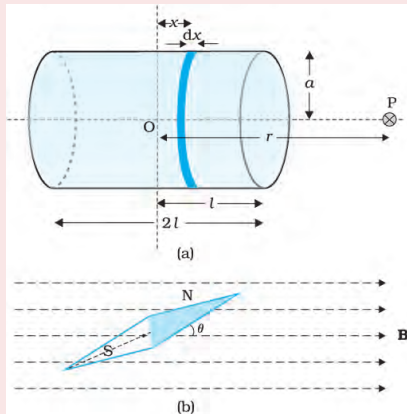
- This integration can be done by trigonometric substitution. But for far axial points **i.e.**, for $r \gg x$, and $r \gg a$ this expression simplifies to

$$B = \frac{\mu_0 l a^2 n}{2} \int_{-l}^l \frac{dx}{[r^2]^{3/2}}$$

$$B = \frac{\mu_0 l a^2 n}{2r^3} \int_{-l}^l dx = \frac{\mu_0 l a^2 n}{2r^3} \times 2l$$

- magnetic moment of solenoid $m = (2nll)(\pi a^2)$

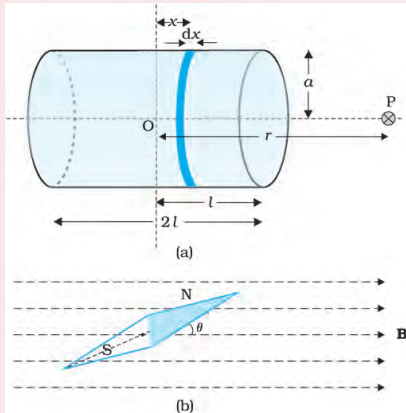
Bar magnet as an equivalent solenoid



$$B = \frac{\mu_0 2m}{4\pi r^3}$$

- This is also the far axial magnetic field of a bar magnet which one may obtain experimentally.
- Thus, a bar magnet and a solenoid produce similar magnetic field.

The dipole in a uniform magnetic field

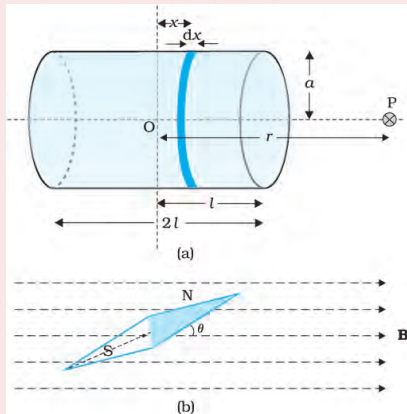


- When we place a magnetic dipole (magnetic needle) of known magnetic moment $\vec{m}$ in the magnetic field $\vec{B}$.
- The **torque in the magnetic dipole** (needle) is

$$\vec{\tau} = \vec{m} \times \vec{B}$$

- In magnitude $\tau = mB \sin \theta$

The dipole in a uniform magnetic field



- An expression for magnetic potential energy U_m can also be obtained on lines similar to electrostatic potential energy.

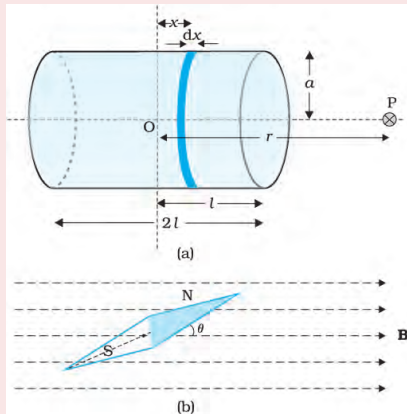
$$U_m = \int \tau(\theta) d\theta$$

$$U_m = \int mB \sin \theta d\theta = -mB \cos \theta$$

$$U_m = -\vec{m} \cdot \vec{B}$$

- We can consider zero potential energy at $\theta = 90^\circ$

The dipole in a uniform magnetic field



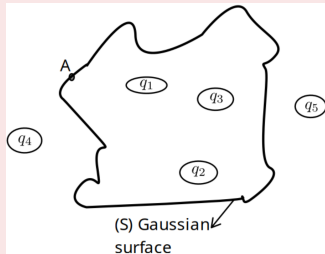
- potential energy is minimum ($U_{min} = -mB$) at $\theta = 0^\circ$ (most stable position) and
- maximum ($U_{max} = +mB$) at $\theta = 180^\circ$ (most unstable position).

The analogy between electric and magnetic dipoles.

	Electrostatics	Magnetism
	$\frac{1}{\epsilon_0}$	μ_0
Dipole moment	$\vec{p}$	$\vec{m}$
Equatorial Field for a short dipole	$\vec{E}_{eq} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p}}{r^3}$	$\vec{B}_{eq} = \frac{\mu_0}{4\pi} \frac{\vec{m}}{r^3}$
Axial Field for a short dipole	$\vec{E}_{axial} = \frac{1}{4\pi\epsilon_0} \frac{2\vec{p}}{r^3}$	$\vec{B}_{axial} = \frac{\mu_0}{4\pi} \frac{2\vec{m}}{r^3}$
External Field: Torque	$\vec{\tau} = \vec{p} \times \vec{E}$	$\vec{\tau} = \vec{m} \times \vec{B}$
External Field: Energy	$U = -\vec{p} \cdot \vec{E}$	$U = -\vec{m} \cdot \vec{B}$

Note: Do all NCERT exercise questions related to this topic in your notebook

MAGNETISM AND GAUSS'S LAW



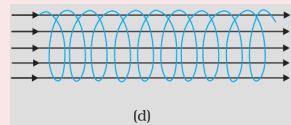
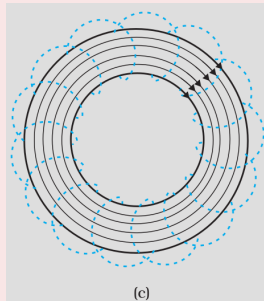
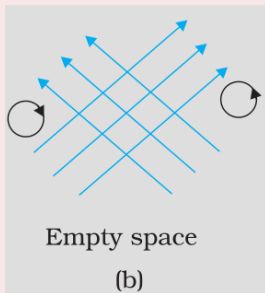
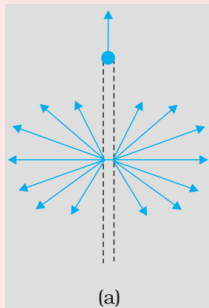
- **Gauss's law for magnetism** is: The **net magnetic flux through any closed surface is zero**.

$$\oint \vec{B} \cdot d\vec{S} = 0$$

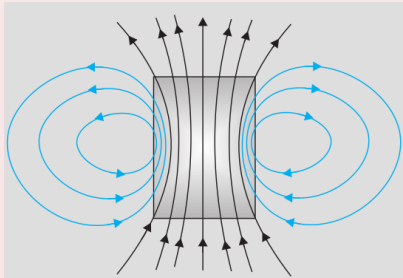
- Number of **magnetic field lines entering** the closed surface is **equal** to the **number of field lines leaving** the surface.
- This law is consistent with the observation that **isolated magnetic poles (monopoles) do not exist**.
- There are **no sources or sinks of magnetic field**.

Example 5.6

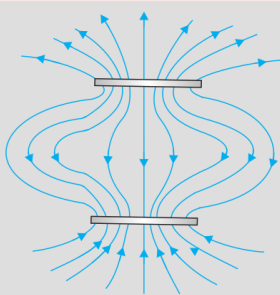
Many of the diagrams given in Fig. 5.7 show magnetic field lines (thick lines in the figure) wrongly. Point out what is wrong with them. Some of them may describe electrostatic field lines correctly. Point out which ones.



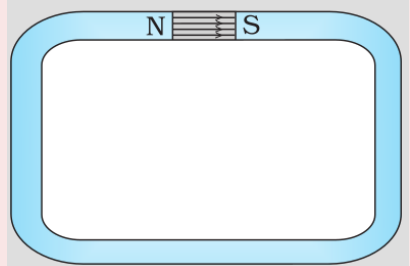
Example 5.6



(e)



(f)



(g)

Example 5.7

- ➊ Magnetic field lines show the direction (at every point) along which a small magnetised needle aligns (at the point). Do the magnetic field lines also represent the lines of force on a moving charged particle at every point?
- ➋ Magnetic field lines can be entirely confined within the core of a toroid, but not within a straight solenoid. Why?
- ➌ If magnetic monopoles existed, how would the Gauss's law of magnetism be modified?
- ➍ Does a bar magnet exert a torque on itself due to its own field? Does one element of a current-carrying wire exert a force on another element of the same wire?
- ➎ Magnetic field arises due to charges in motion. Can a system have magnetic moments even though its net charge is zero?

BASIC TERMS IN MAGNETISM

Magnetisation Or Intensity of Magnetisation ($\vec{M}$) :

- When a magnetic material is placed in an external magnetic field, the atomic dipoles of the material aligned, partially or fully, producing a net magnetic moment ($\vec{m}$).

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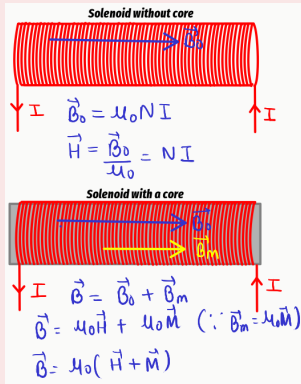
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- S.I. Unit of $\vec{M} = \frac{A \, m^2}{m^3} = A \, m^{-1}$

BASIC TERMS IN MAGNETISM

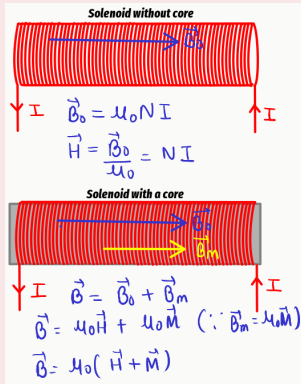
Relation between $\vec{B}$, $\vec{M}$ and $\vec{H}$:



- When a magnetic material is placed in a magnetic field, the material gets magnetised.

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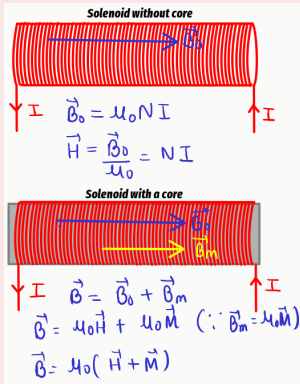
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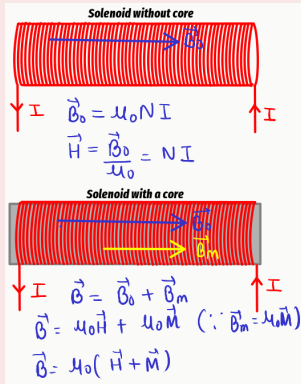


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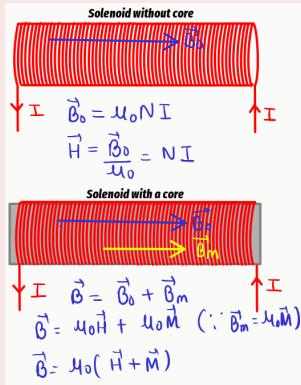
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- χ is called **magnetic susceptibility** it is dimensionless quantity. for vacuum $\chi = 0$

BASIC TERMS IN MAGNETISM

- From eq (4) and (5)

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$$\boxed{\vec{B} = \mu_0 \mu_r \vec{H} = \mu \vec{H}} \dots (6)$$

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$$\boxed{\mu = \mu_0 \mu_r = \mu_0 (1 + \chi)} \dots (7)$$

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Example 5.10

A solenoid has a core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2 A. If the number of turns is 1000 per metre, calculate (a) H , (b) B , (c) M and (d) the magnetising current I_m .

Classification of magnetic materials

- **Depending on** the way the **magnetic moments** of the electrons in the **atoms** of a material **react to an applied magnetic field**, materials can be **classified as diamagnetic, paramagnetic and ferromagnetic materials**.

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Diamagnetic	Paramagnetic	Ferromagnetic
$-1 \leq \chi < 0$	$0 < \chi < \varepsilon$	$\chi \gg 1$
$0 \leq \mu_r < 1$	$1 < \mu_r < 1 + \varepsilon$	$\mu_r \gg 1$
$\mu < \mu_0$	$\mu > \mu_0$	$\mu \gg \mu_0$
	Here ε is a small positive number	

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Classification of magnetic materials

Diamagnetic material :

Characteristics :

- They possess **small and negative susceptibility** ($\chi < 0$).

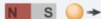
$\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$

No unpaired electrons

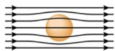
$\mathbf{B} \longrightarrow$



Anti-parallel



Very weakly repelled



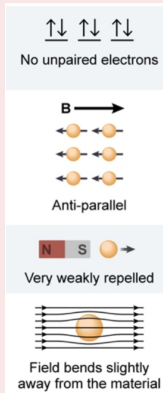
Field bends slightly
away from the material

Classification of magnetic materials

Diamagnetic material :

Characteristics :

- They possess **small and negative susceptibility** ($\chi < 0$).
- The **Magnetic moment of the atoms align** in the **opposite direction** to applied magnetic field.

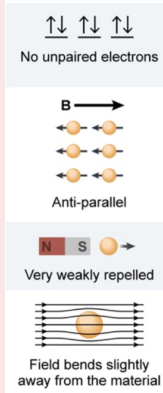


Classification of magnetic materials

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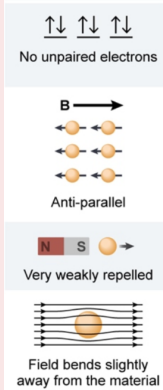


Classification of magnetic materials

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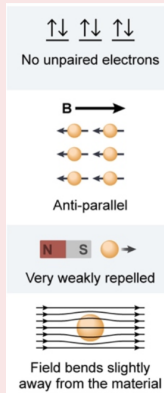
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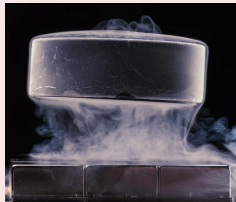
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- They **do not retain the magnetic properties** when the **external field is removed**.

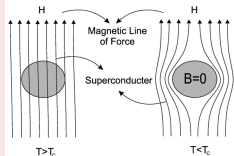
Classification of magnetic materials

Diamagnetic material :



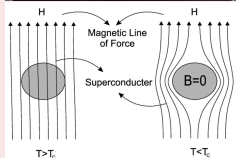
Characteristics :

- **Susceptibility is independent of temperature** of the material.



Classification of magnetic materials

Diamagnetic material :

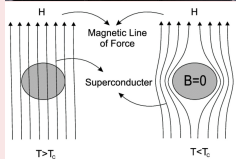


Characteristics :

- **Susceptibility is independent of temperature** of the material.
- **Superconductors** – Superconductors are essentially diamagnetic materials with a volume **susceptibility** of $\chi = -1$.

Classification of magnetic materials

Diamagnetic material :

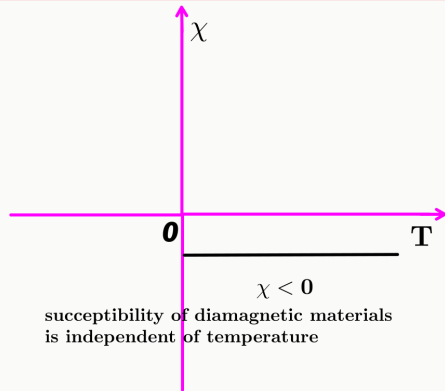
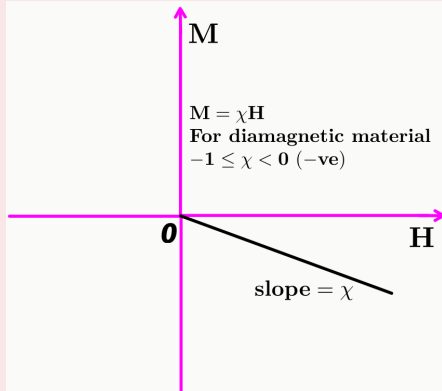


Characteristics :

- **Susceptibility is independent of temperature** of the material.
- **Superconductors** – Superconductors are essentially diamagnetic materials with a volume **susceptibility** of $\chi = -1$. They exhibit complete diamagnetic screening and may be classified as **perfect diamagnets** since they **expel all magnetic fields**.

Classification of magnetic materials

Diamagnetic material :



Classification of magnetic materials

Paramagnetic materials :

- Those **substances which** are **weakly magnetised in the same direction as the applied magnetic field** are called paramagnetic substances.

Classification of magnetic materials

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Classification of magnetic materials

Paramagnetic materials :

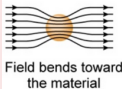
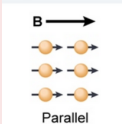
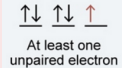
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Classification of magnetic materials

Paramagnetic materials :

Characteristics :

- They possess small and positive susceptibility. $\chi > 0$ and small.

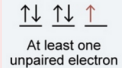


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B →



Parallel



Attracted



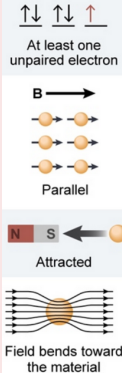
Field bends toward the material

Classification of magnetic materials

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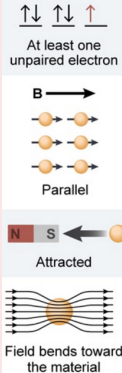


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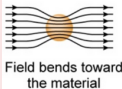
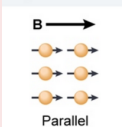
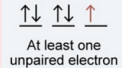
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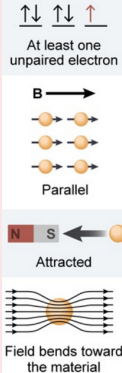


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Classification of magnetic materials

Paramagnetic materials :



At least one unpaired electron

$B \rightarrow$



Parallel



Attracted



Field bends toward the material

Curie's law :

- The **magnetisation** of a paramagnetic material is **inversely proportional** to the absolute temperature T .

Classification of magnetic materials

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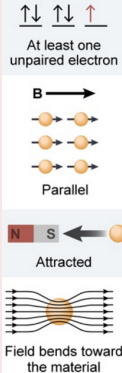
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Classification of magnetic materials

Paramagnetic materials :



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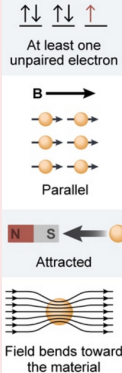
- The **magnetisation** of a paramagnetic material is **inversely proportional** to the absolute temperature T .

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- The **constant C** is called **Curie's constant**.

Classification of magnetic materials

Paramagnetic materials :



Curie's law :

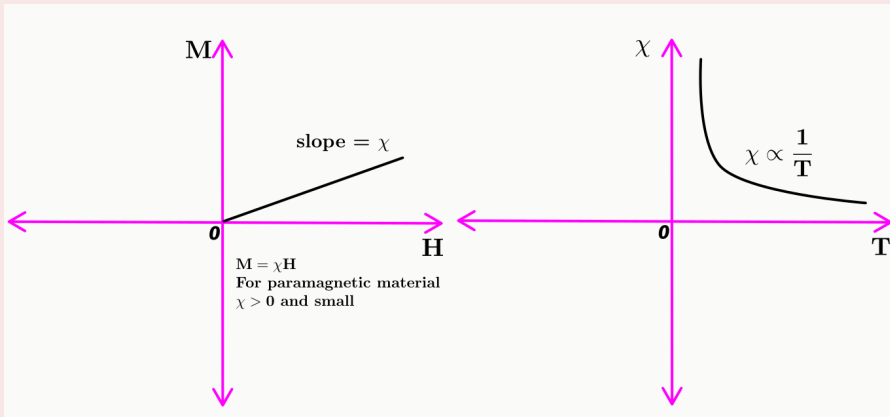
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- The constant C is called Curie's constant.
- Thus, for a paramagnetic material both χ and μ_r depend not only on the material, but also on the sample temperature.

Classification of magnetic materials

Paramagnetic materials :



Classification of magnetic materials

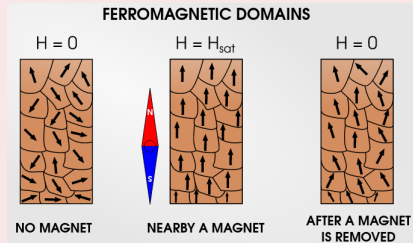
Ferromagnetic materials :

- Those **substances which are strongly magnetised** in the **same direction as that of applied magnetic field** and **retains its magnetic moment** even **after the removal of the applied field** are called ferromagnetic materials.

Classification of magnetic materials

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- **Examples:** Iron, cobalt, nickel



Classification of magnetic materials

Ferromagnetic materials :

- **Explanation:** Ferromagnetism is a special case of paramagnetism.

Classification of magnetic materials

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Characteristics :

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Classification of magnetic materials

Ferromagnetic materials :

- **Hard ferromagnets** : They **retains a strong magnetic field** (makes a good permanent magnet), and is **difficult to demagnetize**. eg. **Alnico (alloy of Al, Ni, and Co)**

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$$\chi = \frac{C}{T - T_c} \quad T > T_c$$

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A ferromagnetic material behaves as a paramagnetic material when temperature is greater than the Curie temperature T_c .

Classification of magnetic materials

Paramagnetic materials :

